

Relevance Score Assignment for Artificial Neural Networks

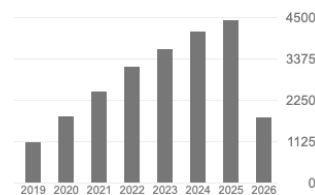
2016

US 20180018553 "Relevance Score Assignment for Artificial Neural Networks "	granted 2024-10-04
CN 107636693 "Relevance Score Assignment for Artificial Neural Networks"	granted 2022-01-11
EP 3271863 "Relevance Score Assignment for Artificial Neural Network"	granted 2021-07-28
JP 6725547 "Relevance Score Assignment for Artificial Neural Networks"	granted 2020-07-22
KR 102130162 "Assignment of Relevance Scores for Artificial Neural Networks"	granted 2020-07-06
CA 2979579 "Relevance Score Assignment for Artificial Neural Networks"	granted 2020-02-18
RU 2703343 "Relevancy Assessment for Artificial Neural Networks"	granted 2019-10-16
BR 112017019821 "Relevance Score Assignment for Artificial Neural Networks "	published 2018-05-15
WO 2016150472 A1 "Relevance score assignment for artificial neural network"	published 2016-09-29

Publications

Summary of Scientific Impact

	All	Since 2021
# Publications	109	83
# Citations	23645	19630
h-index	40	39
i10-index	68	66



per Google Scholar, retrieved on May 30th, 2026.

List of Publications

Journal Articles

- Anders C J, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2026).
"Software for Dataset-wide XAI: From Local Explanations to Global Insights with Zennit, CoRelAy, and ViRelAy".
In: *PLoS ONE* 21(1):e0336683.
<https://github.com/chr5tphr/zennit> | <https://github.com/virelay/corelay> | <https://github.com/virelay/virelay>
- Cantú E D, Wittmann R K, Abdeen O, Wagner P, Samek W, Baier M and **Lapuschkin S** (2026).
"Deep Learning-based Multi Project InP Wafer Simulation towards Unsupervised Surface Defect Detection".
In: *Journal of the Franklin Institute* 363(2):108316
- Yolcu G Ü, Weckbecker M, Wiegand T, Samek W and **Lapuschkin S** (2025).
"Sparse, Efficient and Explainable Data Attribution with DualXDA".
In: *Transactions on Machine Learning Research* qfx81N884A.
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- Dreyer M, Berend J, Labarta T, Vielhaben J, Wiegand T, **Lapuschkin S** and Samek W (2025).
"Mechanistic Understanding and Validation of Large AI Models with SemanticLens".
In: *Nature Machine Intelligence* 1–14.
<https://github.com/jim-berend/semanticlens> | Demo: <https://semanticlens.hhi-research-insights.eu>
- Pahde F, Wiegand T, **Lapuschkin S** and Samek W (2025).
"Ensuring Medical AI Safety: Explainable AI-Driven Detection and Mitigation of Spurious Model Behavior and Associated Data".
In: *Machine Learning* 114(9):206.
<https://github.com/frederikpahde/medical-ai-safety>
- Ma J, Weicken E, Pahde F, Weitz K, **Lapuschkin S**, Samek W and Wiegand T (2025).
"Künstliche Intelligenz auf dem Prüfstand: Anforderungen, Qualitätskriterien und Prüfwerkzeuge für medizinische Anwendungen [Artificial intelligence under scrutiny: requirements, quality criteria, and testing tools for medical applications]".
In: *Bundesgesundheitsblatt – Gesundheitsforschung – Gesundheitsschutz* 68:915-923

7. Storås A M, Dreyer M, Pahde F, **Lapuschkin S**, Samek W, Halvorsen P, de Lange T, Mori Y, Hann A, Berzin T M, Parasa S and Riegler M A (2025).
“Exploring the Clinical Value of Concept-based AI Explanations in Gastrointestinal Disease Detection”.
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8. Weber L, Berend J, Weckbecker M, Binder A, Wiegand T, Samek W and **Lapuschkin S** (2025).
“Efficient and Flexible Neural Network Training through Layer-wise Feedback Propagation”.
In: *Transactions on Machine Learning Research* 9oToxYVOSW.
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9. Hedström A, Bommer P L, Burns T F, **Lapuschkin S**, Samek W and Höhne M-C M (2025).
“Evaluating Interpretable Methods via Geometric Alignment of Functional Distortions”.
In: *Transactions on Machine Learning Research* ukLxqA8zXj.
<https://github.com/annahedstroem/GEF/> | *TMLR Survey Certification*
10. Bley F, **Lapuschkin S**, Samek W and Montavon G (2025).
“Explaining Predictive Uncertainty by Exposing Second-Order Effects”.
In: *Pattern Recognition* 160:111171.
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11. Vielhaben J, **Lapuschkin S**, Montavon G and Samek W (2024).
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In: *Pattern Recognition* 150:110309.
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12. Becker S, Vielhaben J, Ackermann M, Müller K-R, **Lapuschkin S** and Samek W (2024).
“AudioMNIST: Exploring Explainable Artificial Intelligence for Audio Analysis on a Simple Benchmark”.
In: *Journal of the Franklin Institute* 361(1):418–428.
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13. Achibat R, Dreyer M, Eisenbraun I, Bosse S, Wiegand T, Samek W and **Lapuschkin S** (2023).
“From attribution maps to human-understandable explanations through Concept Relevance Propagation”.
In: *Nature Machine Intelligence* 5(9):1006–1019.
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14. Hedström A, Bommer P, Wickstrøm K K, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
“The Meta-Evaluation Problem in Explainable AI: Identifying Reliable Estimators with MetaQuantus”.
In: *Transactions on Machine Learning Research* j3FK00HyfU.
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15. Weber L, **Lapuschkin S**, Binder A and Samek W (2023).
“Beyond Explaining: Opportunities and Challenges of XAI-Based Model Improvement”.
In: *Information Fusion* 92:154–176
16. Hedström A, Weber L, Krakowczyk D G, Bareeva D, Motzkus F, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
“Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond”.
In: *Journal of Machine Learning Research* 24(34):1–11.
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17. Hofmann S M, Beyer F, **Lapuschkin S**, Golterman O, Loeffler M, Müller K-R, Villringer A, Samek W and Witte A V (2022).
“Towards the Interpretability of Deep Learning Models for Multi-modal Neuroimaging: Finding Structural Changes of the Ageing Brain”.
In: *NeuroImage* 261:119504
18. Ma J, Schneider L, **Lapuschkin S**, Achibat R, Duchrau M, Krois J, Schwendicke F and Samek W (2022).
“Towards Trustworthy AI in Dentistry”.
In: *Journal of Dental Research* 00220345221106086
19. Rieckmann A, Dworzynski P, Arras L, **Lapuschkin S**, Samek W, Onyebuchi A A, Rod N H, Ekstrøm C T (2022).
“Causes of Outcome Learning: A Causal Inference-inspired Machine Learning Approach to Disentangling Common Combinations of Potential Causes of a Health Outcome”.
In: *International Journal of Epidemiology* dyac078.
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 In: *ACM Transactions on Computing for Healthcare* 3(2):14:1–27.
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21. Anders C J, Weber L, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2022).
 “Finding and Removing Clever Hans: Using Explanation Methods to Debug and Improve Deep Models”.
 In: *Information Fusion* 77:261–295
22. Sun J, **Lapuschkin S**, Samek W and Binder A (2022).
 “Explain and Improve: LRP-inference Fine-tuning for Image Captioning Models”.
 In: *Information Fusion* 77:233–246
23. Samek W, Montavon G, **Lapuschkin S**, Anders C J, and Müller K-R (2021).
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 In: *Proceedings of the IEEE* 109(3):247–278
24. Yeom S-K, Seegerer P, **Lapuschkin S**, Binder A, Wiedemann S, Müller K-R and Samek W (2021).
 “Pruning by Explaining: A Novel Criterion for Deep Neural Network Pruning”.
 In: *Pattern Recognition* 115:107899.
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25. Aeles J, Horst F, **Lapuschkin S**, Lacourpaille L, and Hug F (2021).
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26. Horst F, Slijepcevic D, Zeppelzauer M, Raberger AM, **Lapuschkin S**, Samek W, Schöllhorn WI, Breiteneder C, and Horsak B (2020).
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 In: *Gait & Posture* 81(S1):159–160
27. Hägele M, Seegerer P, **Lapuschkin S**, Bockmayr M, Samek W, Klauschen F, Müller K-R and Binder A (2020).
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 In: *Scientific Reports* 10:6423
28. Alber M, **Lapuschkin S**, Seegerer P, Hägele M, Schütt K T, Montavon G, Samek W, Müller K-R, Dähne S and Kindermans P-J (2019).
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29. **Lapuschkin S**, Wäldchen S, Binder A, Montavon G, Samek W and Müller K-R (2019).
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 In: *Nature Communications* 10:1069
30. Horst F, **Lapuschkin S**, Samek W, Müller K-R and Schöllhorn W I (2019).
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31. Montavon G, **Lapuschkin S**, Binder A, Samek W and Müller K-R (2017).
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 In: *Pattern Recognition* 65:211–222.
Pattern Recognition Best Paper Award and Pattern Recognition Medal winner
32. Samek W, Binder A, Montavon G, **Lapuschkin S**, and Müller K-R (2017).
 “Evaluating the Visualization of what a Deep Neural Network has Learned”.
 In: *IEEE Transactions on Neural Networks and Learning Systems*
33. Sturm I, **Lapuschkin S**, Samek W and Müller K-R (2016).
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 In: *Journal of Neuroscience Methods* 274:141–145
34. **Lapuschkin S**, Binder A, Montavon G, Müller K-R and Samek W (2016).
 “The Layer-wise Relevance Propagation Toolbox for Artificial Neural Networks”.
 In: *Journal of Machine Learning Research* 17(114):1–5.
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“On Pixel-wise Explanations for Non-Linear Classifier Decisions by Layer-wise Relevance Propagation”.
In: *PLoS ONE* 10(7):e0130140

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1. Ostermann S, Gurgurov D, Baeumel T, Hedderich M A, **Lapuschkin S**, Samek W and Schmitt V (2026).
“From Weights to Activations: Is Steering the Next Frontier of Adaptation?”.
In: *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*
2. Komorowski P, Golimblevskaia E, Achtibat R, Wiegand T, **Lapuschkin S** and Samek W (2026).
“Attribution-Guided Decoding”.
In: *Proceedings of the International Conference on Learning Representations (ICLR)* .
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3. Kahardipraja P, Achtibat R, Wiegand T, Samek W and **Lapuschkin S** (2025).
“The Atlas of In-Context Learning: How Attention Heads Shape In-Context Retrieval Augmentation”.
In: *Advances in Neural Information Processing Systems (NeurIPS)* .
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4. Bareeva D, Höhne M M C, Warnecke A, Pirch L, Müller K-R, Rieck K, **Lapuschkin S** and Bykov K (2025).
“Manipulating Feature Visualizations with Gradient Slingshots”.
In: *Advances in Neural Information Processing Systems (NeurIPS)* .
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5. Sandmann E, **Lapuschkin S** and Samek W (2025).
“Iterative Inference in a Chess-Playing Neural Network”.
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6. Golimblevskaia E, Jain A, Puri B, Ibrahim A, Samek W and **Lapuschkin S** (2025).
“Circuit Insights: Towards Interpretability Beyond Activations”.
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7. Labarta T, Hoang N, Weitz K, Samek W, **Lapuschkin S** and Weber L (2025).
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8. Puri B, Jain A, Golimblevskaia E, Kahardipraja P, Wiegand T, Samek W and **Lapuschkin S** (2025).
“FADE: Why Bad Descriptions Happen to Good Features”.
In: *Findings of the Association for Computational Linguistics (ACL)* 17138–17160.
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9. Naujoks J, Krasowski A, Weckbecker M, Yolcu G Ü, Wiegand T, **Lapuschkin S**, Samek W and Klausen R P (2025).
“Leveraging Influence Functions for Resampling Data in Physics-Informed Neural Networks”.
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10. Erogullari E, **Lapuschkin S**, Samek W and Pahde F (2025).
“Post-Hoc Concept Disentanglement: From Correlated to Isolated Concept Representations”.
In: *Proceedings of the 3rd XAI World Conference* 68–89.
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11. Joseph S, Suresh P, Hufe L, Stevinson E, Graham R, Vadi Y, Bzdok D, **Lapuschkin S**, Sharkey L and Richards A (2025).
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In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops: MIV Workshop* TBA.
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12. Pahde F, Dreyer M, Weckbecker M, Weber L, Anders C J, Wiegand T, Samek W and **Lapuschkin S** (2025).
“Navigating Neural Space: Revisiting Concept Activation Vectors to Overcome Directional Divergence”.
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13. Bareeva D, Yolcu GÜ, Hedström A, Wiegand T, Samek W and **Lapuschkin S** (2024).
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14. Naujoks J R, Krasowski A, Weckbecker M, Wiegand T, **Lapuschkin S**, Samek W and Klausen R P (2024).
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Reproducibility Badge Winner
 15. Kopf L, Bommer P L, Hedström A, **Lapuschkin S**, Höhne M M-C and Bykov K (2024).
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 17. Mekala R R, Pahde F, Baur S, Chandrashekar S, Diep M, Wenzel M A, Wisotzky E L, Yolcu G Ü, **Lapuschkin S**, Ma J, Eisert P, Lindvall M, Porter A and Samek W (2024).
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 18. Achibat R, Hatefi S M V, Dreyer M, Jain A, Wiegand T, **Lapuschkin S**, Samek W (2024).
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 19. Hatefi S M V, Dreyer M, Achibat R, Wiegand T, Samek W and **Lapuschkin S** (2024).
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 21. Tinauer C, Damulina A, Sackl M, Soellradl M, Achibat R, Dreyer M, Pahde F, **Lapuschkin S**, Schmidt R, Ropele S, Samek W, Langkammer C (2024).
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 22. Dreyer M, Pürelku E, Vielhaben J, Samek W, **Lapuschkin S** (2024).
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 23. Bareeva D, Dreyer M, Pahde F, Samek W and **Lapuschkin S** (2024).
 "Reactive Model Correction: Mitigating Harm to Task-Relevant Features via Conditional Bias Suppression".
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 24. Dreyer M, Achibat R, Samek W and **Lapuschkin S** (2024).
 "Understanding the (Extra-)Ordinary: Validating Deep Model Decisions with Prototypical Concept-based Explanations".
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 25. Dreyer M, Pahde F, Anders C J, Samek W and **Lapuschkin S** (2024).
 "From Hope to Safety: Unlearning Biases of Deep Models via Gradient Penalization in Latent Space".

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26. Dawoud K, Samek W, Eisert P, **Lapuschkin S** and Bosse S (2023).
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 27. Frommholz A, Seipel F, **Lapuschkin S**, Samek W and Vielhaben J (2023).
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 28. Hedström A, Weber L, **Lapuschkin S** and Höhne M M-C (2023).
 “Sanity Checks Revisited: An Exploration to Repair the Model Parameter Randomisation Test”.
 In: *NeurIPS 2023 Workshop on XAI (XAI in Action: Past, Present, and Future Applications)* (vVpefYmnsG)
 29. Pahde F, Dreyer M, Samek W and **Lapuschkin S** (2023).
 “Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models”.
 In: *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 596–606. (Green Open Access)
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 30. Binder A, Weber L, **Lapuschkin S**, Montavon G, Müller K-R and Samek W (2023).
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 31. Dreyer M, Achibat R, Wiegand T, Samek W and **Lapuschkin S** (2023).
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 32. Pahde F, Yolcu GÜ, Binder A, Samek W and **Lapuschkin S** (2023).
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 33. Krakowczyk D G, Prasse P, Reich D R, **Lapuschkin S**, Scheffer T, Jäger L A (2023).
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 34. Krakowczyk D G, Reich D R, Prasse P, **Lapuschkin S**, Jäger L A and Scheffer T (2022).
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 38. Goh G S W, **Lapuschkin S**, Weber L, Samek W and Binder A (2021).
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 39. Kohlbrenner M, Bauer A, Nakajima S, Binder A, Samek W, and **Lapuschkin S** (2020).
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41. Anders C J, Neumann D, Marinč T, Samek W, Müller K-R and **Lapuschkin S** (2020).
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42. Sun J, **Lapuschkin S**, Samek W, Zhao Y, Cheung N-M and Binder A (2020).
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43. Alber M, **Lapuschkin S**, Seegerer P, Hägele M, Schütt K T, Montavon G, Samek W, Müller K-R, Dähne S and Kindermans P-J (2018).
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